

$$\textcircled{1} P(Y) = \frac{5}{12}, P(N) = \frac{7}{12}.$$

$$H(Y) = -\frac{5}{12} \log_2 \frac{5}{12} - \frac{7}{12} \log_2 \frac{7}{12} \\ = 0.980$$

$$\textcircled{2} \cdot \text{spots} = N, y = N \Rightarrow H(N) = 0.$$

$$\text{spots} = Y: y: 2N, 5Y.$$

$$P(Y | \text{spots} = Y) = \frac{5}{7}, P(N | \text{spots} = Y) = \frac{2}{7}.$$

$$H(Y | \text{spots} = Y) = -\frac{5}{7} \log_2 \frac{5}{7} - \frac{2}{7} \log_2 \frac{2}{7} \\ = 0.863$$

$$H(y | x_2 = \text{spots}) = H(N) \cdot P(N) + H(Y) \cdot P(Y)$$

$$= 0 \cdot \frac{5}{12} + 0.863 \cdot \frac{7}{12}$$

$$= 0.503.$$

$$IG(\text{spots}) = H(Y) - H(y | x_2 = \text{spots})$$

$$= 0.980 - 0.503$$

$$= 0.477$$

$$\cdot \text{color} = \text{white}: y = 4N, 2Y.$$

$$P(Y | \text{white}) = \frac{1}{3}, P(N | \text{white}) = \frac{2}{3}.$$

$$H(y | \text{white}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3}$$

$$= 0.918$$

color = Brown :  $y = 3N, 3Y$ .

$$P(Y|Brown) = \frac{1}{2}, P(N|Brown) = \frac{1}{2}.$$

$$\begin{aligned} H(y|Brown) &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= 1 \end{aligned}$$

$$\begin{aligned} H(y|color) &= P(white) \cdot H(y|white) + \\ &\quad P(Brown) \cdot H(y|Brown) \\ &= \frac{1}{2} \cdot 0.918 + \frac{1}{2} \cdot 1 \\ &= 0.959 \end{aligned}$$

$$\begin{aligned} IG(color) &= H(y) - H(y|color) \\ &= 0.980 - 0.959 \\ &= 0.021 \end{aligned}$$

• Size: split at 2.5.

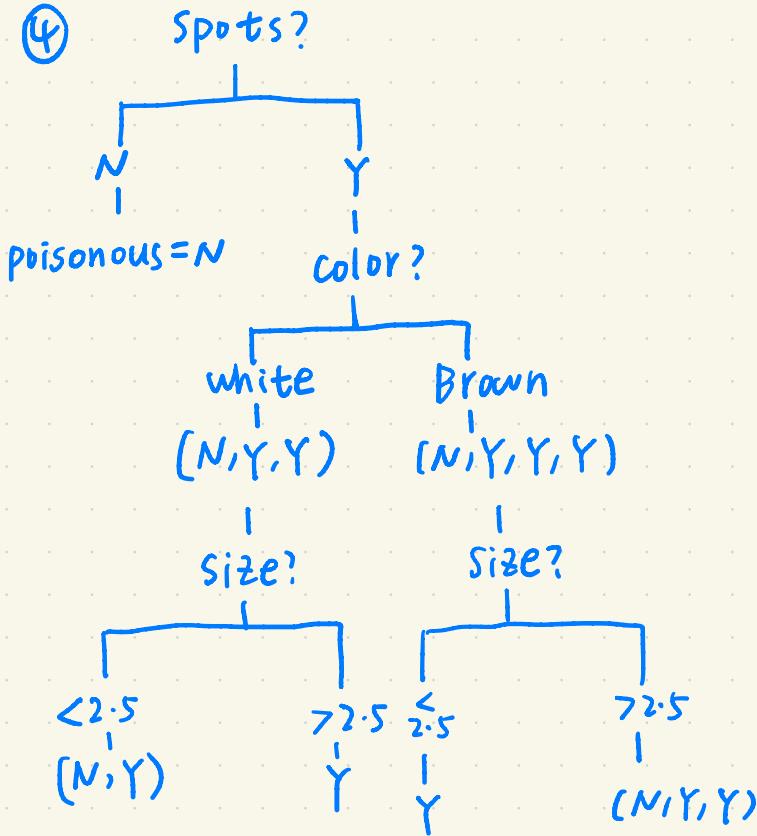
$$H(y|split) = 0.5 \times 0.918 + 0.5 \times 1 = 0.959$$

$$\begin{aligned} IG &= 0.980 - 0.959 \\ &= 0.021. \end{aligned}$$

$\therefore$  first attribute:  $X_2$  (spots).

③ The IG of  $x_2 = 0.477$ .

④



Bonus Questions:

5.1.  $\log_2(4) = 2$ .

2 bits

5.2. It ensures that completely random positions have 0 information and fully conserved positions have 2 bits.

5.3. Using  $\log$  base  $B$  would automatically scale the values between 0 and 1.

5.4.  $\log_{20}$ .